

Multiple Choice (10 marks)

1. Which of the following confirmed values meet the diagnostic threshold for diabetes?
 - (a) fasting blood glucose ≥ 160 mg/dl
 - (b) fasting blood glucose ≥ 140 mg/dl
 - (c) fasting blood glucose ≥ 126 mg/dl
 - (d) random glucose > 180 mg/dl
 - (e) fasting blood glucose ≥ 120 mg/dl
2. Simple tandem repeat polymorphisms in humans are most useful for
 - (a) studying the effects of radiation exposure
 - (b) identifying the presence of bacterial infections
 - (c) solving criminal and paternity cases
 - (d) determining blood type
 - (e) transferring disease resistance factors into bone marrow cells
3. Proprioceptive nerve endings in synovial joints are located in
 - (a) synovial fluid and articular discs.
 - (b) capsule and ligaments.
 - (c) articular cartilage and ligaments.
 - (d) capsule and articular cartilage.
 - (e) synovial membrane and capsule.
4. Severe acute malnutrition in young children is defined as:
 - (a) Weight-for-age Z score < -2 and oedema
 - (b) Height-for-age Z score < -2 or weight-for-height Z score < -2 and oedema
 - (c) Height-for-age Z score < -3 and oedema
 - (d) Weight-for-age Z score < -2 or height-for-age Z score < -2 or oedema
 - (e) Height-for-age Z score < -3 or weight-for-age Z score < -3 and oedema
5. What is the difference between a male and a female catheter?
 - (a) Female catheters are used more frequently than male catheters.
 - (b) Male catheters are bigger than female catheters.
 - (c) Male catheters are more flexible than female catheters.
 - (d) Male catheters are made from a different material than female catheters.
 - (e) Female catheters are longer than male catheters.

True / False (10 marks)

Answer True or False

6. True or False: Grade 2/5 power represents movement of the arm only against resistance.
7. True or False: Having a lower level of education is associated with a decreased risk of Alzheimer's disease.
8. True or False: The maxillary bone has a process that connects to the zygomatic bone, forming the zygomatic arch.
9. True or False: Type I muscle fibers are characterized as red, oxidative, and slow contracting.
10. True or False: The concentration of a food substance is a critical factor in evaluating its safety based on toxicity and dose-response data.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the genetic basis of DiGeorge/Shprintzen syndrome, including the implications of the chromosomal deletion involved and its effects on physiological and developmental outcomes.
12. Discuss the role of ubiquitin in proteolysis, particularly focusing on its attachment to proteins designated for synthesis versus those targeted for catabolism.

End of Paper